

TED (15) – 4021

Reg. No

REVISION – 2015

Signature

FOURTH SEMESTER DIPLOMA MODEL EXAMINATION IN ENGINEERING /
TECHNOLOGY

APPLIED MECHANICS AND STRENGTH OF MATERIALS

(Common for ME, TD and WP)

(Maximum marks : 100)

(Time 3 Hours)

PART – A

(Maximum marks : 10)

Marks

- I Answer the following questions in one or two sentences. Each question carries 2 marks.

1. Define Modulus of rigidity.
2. Define Lateral strain.
3. Define cone of friction..
4. List any 4 types of riveted joints.
5. Define Bending moment

(5 x 2 = 10)

PART – B

- II Answer *any five* of the following. Each question carries 6 marks.

1. A specimen of square section 50 x 50 mm is 200 mm long. It is shortened by 0.3mm under an axial compressive load of 40 kN.

Determine a) Strain. b) Modulus of elasticity.

2. Draw the stress – strain diagram for mild steel and show the salient points.
3. State the laws of solid friction.
4. Explain a). Moment of inertia b). Radius of Gyration
- 5). State the formula for finding the torque transmitted by
a) Solid shaft b) Hollow shaft

6. Explain 1. Hoop stress 2. Longitudinal stress in thin cylinders
7. List the assumptions of Euler's equation for finding crippling load

(5 x 6 = 30)

PART – C

UNIT - 1

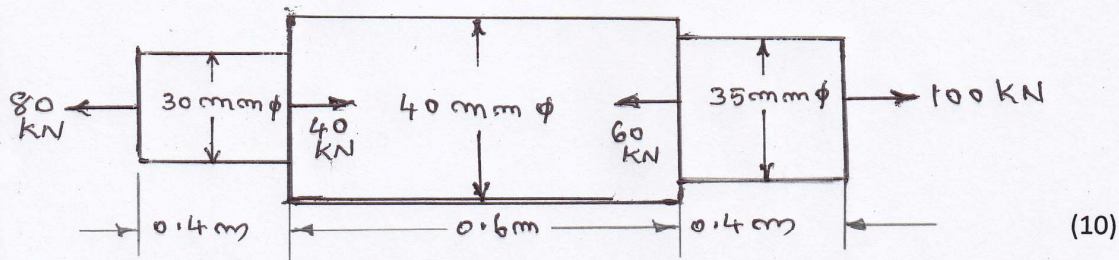
- III 1) In a railway track, new rails are laid at a temperature of 12°C with 3 mm gap at each end. The rails were 10 m long. During the summer, the temperature rose to 45°C . Determine the intensity and kind of stress developed.

Take Coefficient of thermal expansion as $12 \times 10^{-6}/^{\circ}\text{C}$ and Modulus of elasticity as $2 \times 10^5 \text{ N/mm}^2$. (10)

- 2) Explain a) Ultimate stress b). Working stress c). Factor of safety (5)

OR

- IV 1. A bar of varying cross section is loaded as shown in figure. Determine the
a) stress on each section b) Total elongation



2. Find an expression for thermal stress induced in a bar of uniform cross section, when its thermal expansion is completely prevented (5)

UNIT – 2

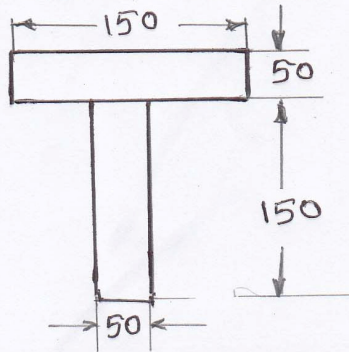
- V 1. A body weighs 500 N is dragged up on a plane inclined at an angle 30° to the horizontal. A force of 400 N inclined at 20° with the plane can just move the body up the plane.

- a). Draw the diagram indicating all the forces.
b). Find the Normal reaction.
c). Find the coefficient of friction. (10)

2. State a) parallel axis theorem b) Perpendicular axis theorem (5)

OR

- VI 1. Find the moment of inertia of T section shown in figure, about X - X axis passing through the centre of gravity of the section.



(All dimensions are in mm) (10)

2. Define a) Sliding friction b) Rolling friction c). Pivot friction (5)

UNIT – 3

- VII 1. 1. a) A solid shaft is to transmit 350 kW at 100 rpm. If the shear stress of shaft material is not to exceed 90 N/mm^2 , find the diameter of the solid shaft.
b) The above solid shaft is replaced by a hollow shaft of same material and length. Its internal diameter is $2/3$ of external diameter. Calculate the percentage of material saved. (10)
2. Illustrate and explain the process and procedure of Caulking and Fullering (5)

OR

- VIII 1. A boiler shell is to be made of 12 mm thick plate having limiting tensile stress of 100 N/mm^2 . Determine the maximum permissible diameter of the shell to withstand a steam pressure of 1.2 N/mm^2 . when
(a) Efficiency of longitudinal joint is 75%
(b) Efficiency of Efficiency of Circumferential joint is 35% (10)
2. List the advantages and disadvantages of welded joint over the riveted joint. (5)

UNIT – 4

- IX 1. A beam of uniform rectangular section 175 mm wide and 250 mm deep is simply supported at its ends. It carries a uniformly distributed load of 7.5 kN/m over the

total length of 4 meter. Modulus of elasticity of beam is $1.2 \times 10^4 \text{ N/mm}^2$.

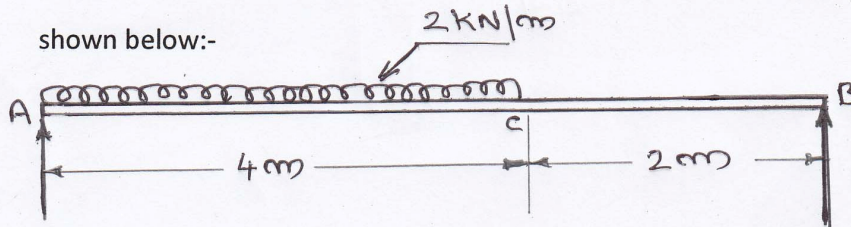
Find a) Slope at the supports

b) Maximum deflection (10)

2. State (a) Euler's formula (b) Rankine's formula for columns (5)

OR

X 1. . Draw the shear force and Bending Moment diagrams for the system of forces shown below:-



(10)

2. (a) Distinguish between long column and short column

(b) Distinguish between Column and strut (5)